Theory-Based User Modeling for Intelligent Tutoring Systems via Grounded Transformers

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Abstract. The student model serves as a cornerstone of Intelligent Tutoring Systems. In many instances, this modeling relies on Knowledge Tracing approaches, which represent the learner's state of mastery across specific skills or knowledge components. However, these traditional approaches are subject to significant limitations. First, they lack pedagogical interpretability, as they are not anchored to theoretical frameworks derived from cognitive or educational principles. Furthermore, they frequently lack longitudinal context, as instructional decisions are typically based on isolated knowledge states without accounting for the historical trajectory that led there. To overcome these limitations, we propose a new approach based on Grounded Transformers that supports instructional decisions grounded in learning theories and individual learning histories, thus providing a theoretically sound foundation for TEL.

Keywords: deep knowledge tracing · theory-based interpretability · learning theories · intelligent tutoring systems · user modeling · grounded transformers

1 Introduction

Knowledge Tracing (KT) [6] is a foundational problem in Intelligent Tutoring Systems (ITS) and Learning Analytics [15] aimed at estimating the evolution of student knowledge as they interact with a sequence of educational items. Accurate mastery tracking enables data-driven personalisation, yet current approaches are caught in a dichotomy between interpretable but rigid probabilistic models and high-accuracy but opaque Deep Knowledge Tracing (DKT) architectures [1,16]. While recent efforts have sought to bridge this gap, they generally do not aim for interpretability grounded in pedagogical principles [2,17]. In a high-stakes domain such as education, where the Artificial Intelligence Act (EU) 2024/1689 explicitly classifies AI systems that assess or monitor learners as high-risk, this lack of transparency raises critical concerns about accountability and trust [10].

To address these limitations, we introduce gTransformer, a Transformer-based model that bridges the gap between deep learning performance and pedagogical interpretability through *representational grounding*: a methodology that explicitly anchors the model's latent representations to the theoretically meaningful concepts of an intrinsically interpretable reference model. By grounding KT in Bayesian Knowledge Tracing (BKT) [6] parameters, gTransformer provides instructional decisions that are both theoretically justifiable and sensitive to each student's longitudinal learning context. This work makes three primary contributions:

– *Grounded Architecture*: We introduce gTransformer, which implements *representational grounding* to bridge deep learning expressiveness and theoretical alignment.
– *Performance–Interpretability Trade-Off*: We demonstrate that grounded models achieve state-of-the-art predictive performance (+19.9% AUC over traditional BKT) while incurring a minimal interpretability cost (3.9%) relative to non-interpretable configurations.
– *Context-Aware Personalisation*: By encoding longitudinal interaction histories into context-aware BKT parameters, gTransformer enables differentiated diagnostics based on individual learning trajectories.

These contributions are structured around three research questions: (RQ1) What is the trade-off between predictive performance and interpretability? (RQ2) How can we verify that the model's internal representations are genuinely grounded in pedagogical theory? (RQ3) How does the longitudinal learning context enhance student-centred personalisation?

2 Background

2.1 Interpretability–Performance Dichotomy

The development of KT has historically been characterised by a dichotomy between two paradigms [1]. Traditional probabilistic models, such as BKT [6] and Factor Analysis Models, are intrinsically interpretable: they model learning through explicit parameters aligned with pedagogical and psychometric principles. However, their simplifying assumptions limit predictive power, causing them to struggle with the complex, non-linear dependencies in real-world student behaviour [11].

DKT models [14], which leverage deep learning architectures from Recurrent Neural Networks to Transformers, achieve state-of-the-art predictive performance. However, this gain has come at the cost of model transparency. Extensive research has sought to bridge this gap through post hoc and ante hoc interpretability methods [2]. While ante hoc approaches integrating IRT parameters [18,4] represent meaningful advances, they frequently produce outputs that remain disconnected from practitioners' domain knowledge. Educators require models that track mastery through concepts explicitly grounded in established frameworks, such as the initial mastery and learn rates postulated by BKT.

2.2 Bayesian Knowledge Tracing as Theoretical Framework

We select BKT [6] as our reference theoretical framework. BKT models the learning process as a Hidden Markov process governed by four interpretable parameters: Initial Mastery ($P(\mathrm{L}_0)$), the probability that a student knows a concept prior to practice; Learn Rate ($P(\mathrm{T})$), the probability of transitioning from the unlearned to the learned state; Guess ($P(\mathrm{G})$), the probability of a correct response despite lack of mastery; and Slip ($P(\mathrm{S})$), the probability of an incorrect response despite mastery. These parameters enable a recursive, causally transparent estimation of student mastery that is directly interpretable by educators.

Standard BKT suffers from a significant limitation: it operates at the population level, estimating a single parameter set for all students on a given skill [19]. gTransformer addresses this by leveraging Transformer attention mechanisms to infer student-specific parameters from individual interaction histories.

2.3 Informed Machine Learning

We situate gTransformer within the Informed Machine Learning (IML) framework [17], which categorises the integration of domain knowledge by its source, representation, and the stage of the machine learning pipeline where it is incorporated. The general IML loss formulation is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{TRN}}(Y_{\mathrm{true}}, Y_{\mathrm{pred}}) + \gamma \mathcal{L}_{\mathrm{THEORY}}(Y_{\mathrm{pred}}) \tag{1}$$

where $\mathcal{L}_{\mathrm{TRN}}$ measures supervised error and $\mathcal{L}_{\mathrm{THEORY}}$ penalises violations of theoretical constraints, weighted by γ. While most IML approaches use prior knowledge to maximise predictive accuracy, gTransformer aims to achieve theory-based interpretability while maintaining competitive performance.

3 The gTransformer Model

3.1 Architecture

Figure 1 illustrates the information flow through gTransformer, mapped to the IML taxonomic dimensions. Student interaction histories are augmented with BKT theoretical priors ($P(\mathrm{L}_0)$, $P(\mathrm{T})$ per skill), encoding prior knowledge directly into the model's input representations. These enriched sequences are processed through an attention-based encoder–decoder Transformer backbone that captures long-range temporal dependencies. The key architectural contribution is a projection module that maps the final context vectors $\mathbf{z}_t$ to student-specific BKT parameter estimates (p_{L_0}, p_{T}), which are then passed through BKT forward logic to generate interpretable mastery predictions.

3.2 Multi-Objective Loss Function

The model is trained with a multi-objective loss function that balances predictive accuracy with pedagogical grounding:

$$\mathcal{L} = \lambda_{\text{sup}}\mathcal{L}_{\text{sup}} + \lambda_{\text{ref}}\mathcal{L}_{\text{ref}} + \lambda_{\text{probe}}\mathcal{L}_{\text{probe}} \tag{2}$$

where $\mathcal{L}_{\text{sup}}$ is the binary cross-entropy on supervised predictions, $\mathcal{L}_{\text{ref}}$ is the cross-entropy on interpretable BKT-based predictions to anchor outputs to the reference model, and $\mathcal{L}_{\text{probe}}$ is an auxiliary loss that encourages latent representations to encode BKT parameters in a linearly recoverable manner. The hyperparameters λ_{sup}, λ_{ref}, λ_{probe} govern the trade-off between performance and interpretability.

4 Experimental Evaluation

4.1 Datasets, Baselines, and Methodology

We evaluate gTransformer on five benchmark datasets with explicit knowledge component mappings: ASSISTments 2009 (AS2009, 4,217 students, 123 skills), ASSISTments 2015 (AS2015, 19,917 students, 100 skills), Algebra 2005 (AL2005, 574 students, 112 skills), Bridge to Algebra 2006 (BDG2006, 1,146 students, 493 skills), and NeurIPS 2020 Education Challenge (NIPS34, 4,918 students, 57 skills). We follow a standard five-fold cross-validation with an 80/20 train/test split, measuring performance via AUC using the question-level mean late-fusion protocol [12]. Baselines include DKT [14], DKVMN [20], SAKT [13], SAINT [5], AKT [7], and ATKT [8].

To quantify the performance–interpretability trade-off (RQ1), we define three metrics. Let p_{sup1} be ablated (non-grounded) supervised predictions, p_{sup2} grounded supervised predictions, p_{ref} interpretable BKT-based predictions, and p_{bkt} classical BKT predictions:

$$\mathcal{I}_{\text{drop}} = p_{\text{sup1}} - p_{\text{sup2}} \quad \text{(cost of grounding on neural capacity)} \tag{3}$$

$$\mathcal{I}_{\text{cost}} = p_{\text{sup2}} - p_{\text{ref}} \quad \text{(cost of switching to interpretable logic)} \tag{4}$$

$$\mathcal{I}_{\text{gain}} = p_{\text{ref}} - p_{\text{bkt}} \quad \text{(advantage over population-level BKT)} \tag{5}$$

To verify grounded interpretability (RQ2), we implement a triple-validation methodology: (H2.1) Structural Alignment, assessed via diagnostic probing [9,3] measuring the linear recoverability of BKT constructs from hidden states; (H2.2) Semantic Alignment, assessed via Spearman's rank correlation (ρ) between grounded parameters and theoretical priors; and (H2.3) Functional Alignment, assessed via a composite confidence metric quantifying the reliability of interpretable predictions as substitutes for supervised ones.

Table 1. Predictive performance (AUC) comparison. Bold indicates best performance per dataset.

Model	AS2009	AS2015	AL2005	BDG2006	NIPS34
AKT	0.7825	0.7081	**0.8306**	**0.8208**	**0.8033**
ATKT	0.7715	**0.7810**	0.7995	0.7889	0.7665
DKT	0.7528	0.7031	0.8149	0.8015	0.7689
DKVMN	0.7440	0.7013	0.8054	0.7983	0.7673
SAKT	0.7263	0.6947	0.7880	0.7740	0.7517
SAINT	0.6921	0.6836	0.7775	0.7781	0.7873
gTransformer	**0.7831**	0.7078	0.8240	0.8148	0.8006

Table 2. AUC–interpretability trade-offs. $\mathcal{I}_{\text{drop}}$ = cost of grounding; $\mathcal{I}_{\text{cost}}$ = interpretability cost; $\mathcal{I}_{\text{gain}}$ = gain over BKT.

Dataset	p_{sup1}	p_{sup2}	p_{ref}	p_{bkt}	$\mathcal{I}_{\text{drop}}$	$\mathcal{I}_{\text{cost}}$	$\mathcal{I}_{\text{gain}}$
AS2009	0.7831	0.7814	0.7436	0.6097	0.2%	4.8%	22.0%
AS2015	0.7078	0.7070	0.6940	—	0.1%	1.8%	—
AL2005	0.8240	0.8237	0.7800	0.7215	0.0%	5.3%	8.1%
BDG2006	0.8148	0.8107	0.7810	0.6756	0.5%	3.6%	15.6%
NIPS34	0.8006	0.7987	0.7666	0.5729	0.2%	4.0%	33.8%
Mean	0.7861	0.7843	0.7530	0.6449	0.2%	3.9%	19.9%

4.2 RQ1: Performance–Interpretability Trade-Off

Table 1 shows the comparative performance of gTransformer against baselines in the ablated (non-grounded) configuration. gTransformer achieves the best AUC on AS2009, ranks second on AL2005, BDG2006, and NIPS34, and third on AS2015, demonstrating that the architecture is competitive with the state of the art.

Table 2 quantifies the trade-off. The interpretability drop ($\mathcal{I}_{\text{drop}}$) is negligible on average (0.2%), demonstrating that grounding constraints do not compromise the model's expressive capacity. The interpretability cost ($\mathcal{I}_{\text{cost}}$), i.e., the performance reduction when constraining predictions to BKT logic, averages 3.9% across datasets. Against population-level BKT, interpretable gTransformer predictions show a consistent advantage of 19.9% on average. These results establish that pedagogical interpretability can be achieved at a minimal performance cost.

4.3 RQ2: Grounded Interpretability

We validate grounded interpretability using AS2009 as the reference dataset.

Structural Alignment (H2.1). Using diagnostic probing, we assess whether BKT constructs are linearly recoverable from final hidden states. Initial Mastery achieves fidelity $R^2 = 0.549$ ($r = 0.743$) with selectivity $\Delta R^2 = 0.619$, exceeding the 0.5

threshold that distinguishes genuine structural encoding from spurious correlations. Learn Rate achieves $R^2 = 0.515$ ($\Delta R^2 = 0.570$). These results confirm that BKT constructs are preserved as primary organising principles in the latent space.

Semantic Alignment (H2.2). Initial Mastery shows weaker monotonic alignment with theoretical priors ($\rho = 0.360$, MAE $= 0.168$), reflecting strong student-specific individualisation. Learn Rate demonstrates moderate alignment ($\rho = 0.544$, MAE $= 0.092$), preserving the pedagogical understanding of practice effects. Both parameters remain within pedagogical bounds, confirming that the model neither abandons nor mechanically reproduces theoretical priors.

Functional Alignment (H2.3). A composite confidence metric shows that 89.2% of student–skill pairs achieve medium or high confidence in interpretable predictions, indicating that theory-grounded outputs provide actionable diagnostic value.

4.4 RQ3: Context-Aware Personalisation

To demonstrate gTransformer's capacity for context-aware personalisation, we cluster students in AS2009 along two dimensions—initial mastery (p_{L_0}) and learn rate (p_{T})—identifying four learning situations: (1) low initial mastery/low learn rate, (2) low initial mastery/high learn rate, (3) high initial mastery/low learn rate, and (4) high initial mastery/high learn rate.

Figure 2 shows mastery trajectories for students who provide identical response sequences to the same questions but belong to different learning situations. Classical BKT predictions (dotted grey lines) overlap perfectly, reflecting its Markovian constraint: given the same response sequence, BKT produces identical probability estimates regardless of learning history. gTransformer predictions (solid coloured lines) diverge based on the full interaction history, demonstrating the model's capacity to differentiate students not by what they answered, but by how they have learned across their entire trajectory.

5 Discussion: Towards Mindful TEL

5.1 Theory-Grounded Instructional Decisions

The results demonstrate that gTransformer addresses the accuracy–interpretability trade-off in KT by providing pedagogically grounded parameters at a minimal performance cost. This is directly relevant to the ECTEL 2026 theme of Mindful TEL: the model embodies the principle that learning technologies should be shaped with intention, ensuring that the cognitive engine of an ITS is anchored in established educational theory rather than opaque data-driven heuristics.

By operationalising BKT parameters (P_{L_0}, P_{T}) derived from deep learning context, gTransformer provides educators with the diagnostic handles through which an ITS can perform mindful instruction. Rather than acting as a black box that issues recommendations without justification, the system grounds its decisions in constructs that practitioners recognise and can validate. This aligns with broader EU imperatives for explainable, accountable AI in education [10], and responds directly to the concerns raised by the Artificial Intelligence Act's designation of educational AI as high-risk.

5.2 Beyond Static Labelling: Context-Aware Adaptation

A central contribution of this work is the shift from population-level labelling to situation-based diagnosis. Our results (RQ3) demonstrate that gTransformer's context-aware predictions provide a higher-resolution understanding of the learning journey than population-level models. By encoding each student's longitudinal history into personalisable parameters, the model acknowledges that students are dynamic learners whose instructional needs evolve continuously.

This enables educators to identify, for instance, students with high prior knowledge but low learn rate—suggesting disengagement or instructional mismatch—and adapt the pacing accordingly before significant learning gaps emerge. Conversely, students with high prior knowledge and high learn rate can be identified for accelerated progression to prevent unnecessary remediation. This intentional design addresses essential concerns about accountability and pedagogical integrity in automated instruction.

5.3 Pedagogical Integrity and Human Agency

The triple-validation methodology (RQ2) establishes that the model's internal representations genuinely internalise pedagogical theory rather than merely correlating with it. Structural alignment confirms BKT constructs as primary organising principles; semantic alignment confirms that grounded parameters retain their pedagogical semantics; and functional alignment quantifies the confidence with which educators can trust interpretable predictions. Together, these results support the claim that gTransformer is not merely a high-performance prediction system, but a theory-grounded diagnostic tool that can serve as the cognitive foundation of responsible, human-centred ITS.

6 Conclusions

This work introduces gTransformer, a Transformer-based model that bridges deep learning performance and pedagogical interpretability through *representational grounding*. Our results validate the approach across three dimensions. Interpretable gTransformer predictions consistently surpass classical BKT with an average gain of 19.9% AUC, at a low interpretability cost of 3.9% relative to black-box architectures (RQ1). The model internalises BKT constructs as its primary organising principle in latent representations, confirmed by structural, semantic, and functional alignment (RQ2). gTransformer enables context-aware personalisation by capturing longitudinal learning patterns that Markovian models cannot represent (RQ3).

As a foundation for TEL, gTransformer demonstrates that pedagogically justifiable student modeling is achievable without sacrificing the predictive power needed for effective adaptive instruction. The *representational grounding* methodology is domain-independent and may be extended to other theoretical frameworks beyond BKT or to other scientific fields where established theoretical models coexist with high-volume empirical data. Future research will investigate the integration of guess and slip parameters for finer-grained personalisation and the generalisation of gTransformer to alternative educational theories.

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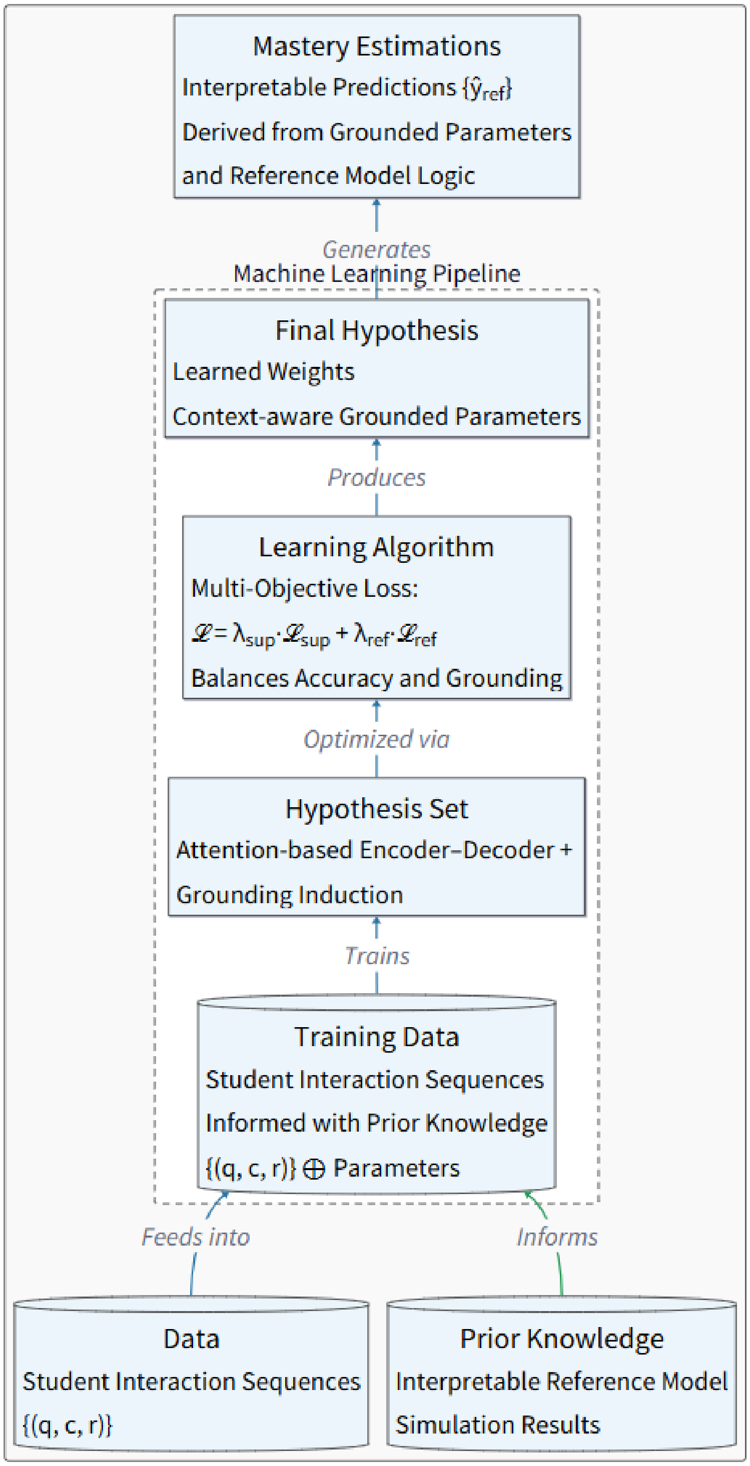

Fig. 1. Dimensions of Informed Machine Learning in gTransformer. The model integrates BKT theoretical priors at the input stage and applies a multi-objective loss to anchor internal representations and outputs to pedagogically meaningful constructs.

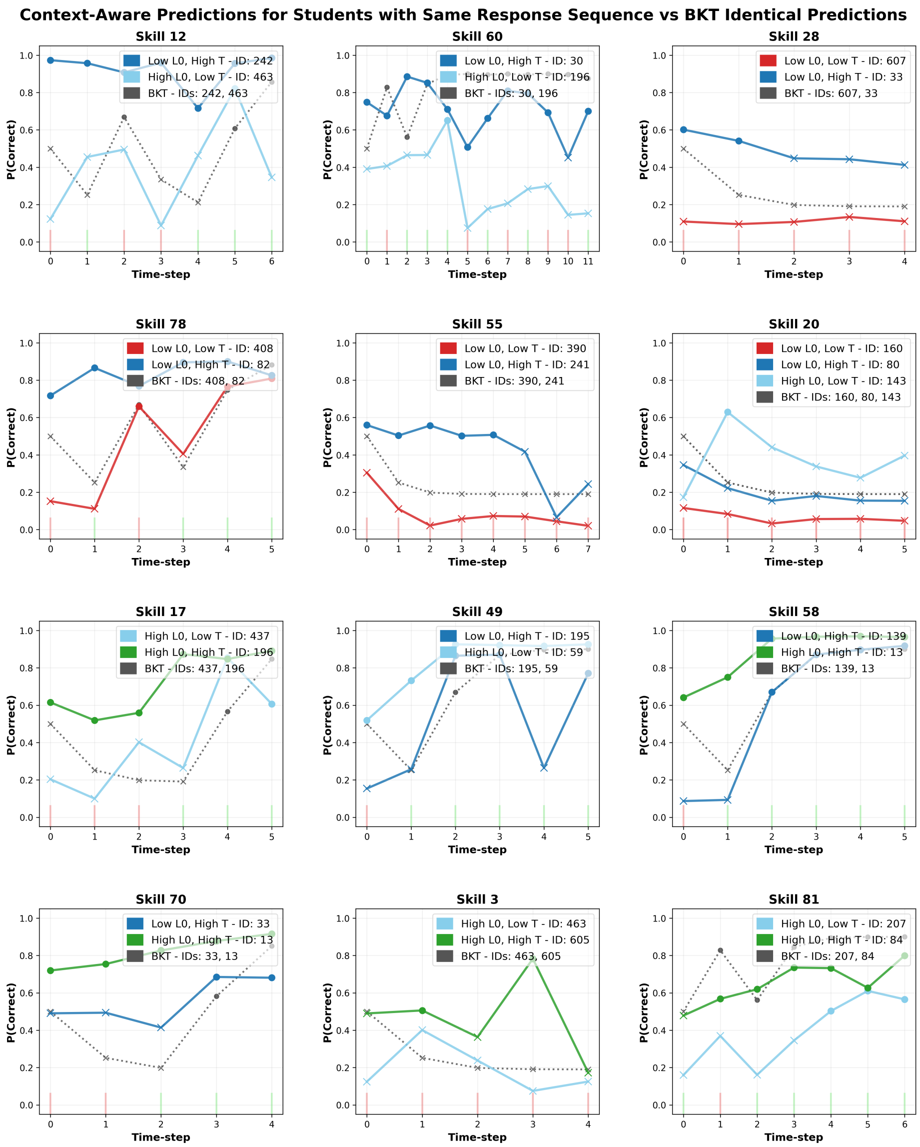

Fig. 2. Context-aware prediction mosaic. Each panel shows students with identical response sequences (green/red bars for correct/incorrect responses) but different learning situations. BKT predictions (dotted grey) overlap due to Markovian constraints; gTransformer predictions (solid coloured) diverge based on historical parameters (p_{L_0}, p_{T}). Circles indicate predicted correct responses; crosses indicate predicted failures.